

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech. (CCT)(CBCS) Degree Examinations
 January 2023

22CCT101- CONSTRUCTION PLANNING AND CONTROL MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

- Note:** 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
 2) Use of **Normal distribution chart** is allowed

Unit – I

Marks BT CO PO

- | | | | | | |
|-------|---|---|-----|---|-----|
| 1. a) | What is line and staff organization? Highlight pros and cons of line and staff organization. | 8 | L*2 | 1 | 1 |
| b) | Give the definition of following terms referring to network analysis
i) Earliest start time & Earliest completion time
ii) Latest start time & Latest completion time
iii) Backward pass & Forward pass | 6 | L1 | 1 | 2 |
| c) | Explain i) Cost breakdown structure and ii) Activity and event | 6 | L2 | 1 | 2 |
| 2. a) | What is work breakdown structure? Explain with the help of flow chart and list the advantages. | 8 | L2 | 1 | 1,3 |
| b) | What is a bar chart? List the advantages and disadvantages of bar chart. | 6 | L1 | 1 | 1 |
| c) | Draw the network representing the following activity relationship using minimum number of dummies.
A, B and C are the initial activities
Q, M and R the finishing activities.
H precedes N but follows D
E follows D but precedes N
F and L follow K and B
F precedes M
Q follows P, L and N
L precedes R but follows C
D precedes G
K and D follow A
F precedes P but follows G | 6 | L3 | 1 | 3 |

Unit – II

- | | | | | | |
|-------|---|---|----|---|---|
| 3. a) | Differentiate between CPM and PERT | 6 | L2 | 2 | 2 |
| b) | Draw the logic network and perform the CPM calculations for the schedule shown below. | | | | |

Activity	A	B	C	D	E	F	G
Immediate Preceding Activity	-	A	A	B	B,C	C	D,E,F
Duration (Days)	5	8	6	9	6	3	1

14 L3 2 3

- | | | | | | |
|-------|---|---|----|---|-----|
| 4. a) | Explain the following with relevant sketches and equations.
i) Total float ii) Free float | 6 | L2 | 2 | 1,3 |
| b) | The following table gives the activities in a construction project and other relevant information | | | | |

Activity	1-2	1-3	2-3	2-4	3-4	4-5	4-6	5-7	6-7
Duration (Days)	20	25	10	12	5	10	5	10	8

Draw the network, Determine the critical path, Find out Earliest times, Latest times, Total float and Free float for each activity

14 L3 2 3

Unit – III

5. a) Explain the following relating to PERT

i) Expected time of completion (t_E)

ii) Probability factor (Z)

iii) Normal distribution curve

6 L2 3 2

- b) A Project with the following activities and durations are given below.

Activity	Time estimates (in Days)		
	t_o	t_m	t_p
1-2	2	5	8
2-3	8	11	20
3-4	0	0	0
2-4	4	7	16
2-5	4	9	20
4-6	7	10	13
5-6	3	7	17
3-7	3	5	13
6-7	2	3	10
7-8	2	4	6

i) Draw a network diagram of the project.

ii) Calculate project duration and variance of all activities

iii) Determine the probability of completing the project in 40 days

14 L3 3 3

6. a) Explain Line of Balance Scheduling technique in project management

8 L2 3 1,3

- b) Highlight the relevance of schedule updating in construction projects?

6 L2 3 1

- c) Write short note on following by highlighting their relevance in project management.

i) Resource Allocation

ii) Resource Leveling

6 L2 3 1

Unit – IV

7. a) For the data given in the table below, draw the network, crash systematically the activities and determine the optimal project duration and cost. Indirect (overhead) cost is Rs 70 per day.

Activity	Normal Duration (Days)	Crash Duration (Days)	Normal Cost (Rs)	Crash Cost (Rs)
1-2	8	6	100	200
1-3	4	2	150	350
2-4	2	1	50	920
2-5	10	5	100	400
3-4	5	1	100	200

20 L5 4 5

8. a) Solve the following L P Problem by graphical method.

Maximize $Z = 2X_1 + X_2$ subject to $X_1 + 2X_2 \leq 10$ $X_1 + X_2 \leq 6$ $X_1 - X_2 \leq 2$ $X_1 - 2X_2 \leq 1$ $X_1 \geq 0$ and $X_2 \geq 0$

8 L3 4 3

- b) Solve following LPP by simplex method
 Maximize $Z = 12X_1 + 16X_2$
 Subject to $10X_1 + 20X_2 \leq 120$
 $8X_1 + 8X_2 \leq 80$
 $X_1, X_2 \geq 0$

12 L5 4 5

Unit – V

9. a) Explain the objectives and functions of material management
 b) Define and explain the following terms.
 i) A B C Analysis of inventory control
 ii) Economic order quantity

10 L2 5 1,3

10 L2 5 1

10. a) Jay deed trading company buys 6,000 units of an item every year with a unit cost of Rs 300.00. It costs Rs 1250.00 to process an order and arrange delivery, while interest and storage cost amounts to Rs 60.00 a year for each unit held. What is the best ordering policy for the item?
 b) Highlight the objectives and functions of store management

10 L2 5 2

10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
